

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

FT-5005(3)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Operation Research and Supply Chain

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain role of Banks in Indian Economy. Discuss lead time variation and safety stock in India.
2. Differentiate between Just In Time (JIT) and Vendor Managed Inventory (VMI).
3. What do you understand by queue? Give some important applications of queueing theory.
4. Define Supply Chain Management. What are the Forecasting Models in SCM?
5. Explain in brief
 - i) ERP
 - ii) MRP

FT-5005(3)-CBGS

PTO

[2]

6. Explain the flow of information, material and money in a supply chain management.
7. What are the basic assumptions made for formulating linear programming problem?

OR

Solve the following problem using simplex method

$$\text{Minimize } z = 8x_1 + 4x_2 + 2x_3$$

$$\text{Subjected to } 4x_1 + 2x_2 + x_3 \leq 8$$

$$3x_1 + 2x_3 \leq 10$$

$$x_1 + x_2 + x_3 = 4$$

$$x_1, x_2, x_3 \geq 0$$

8. Write short notes on any two of the following:
 - a) Non linear optimization problem
 - b) Just in time (JIT)
 - c) Hurwicz criteria of decision analysis
